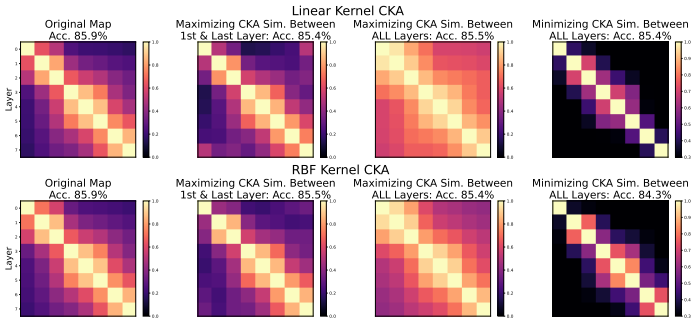

Figure 5: Original Map is the CKA map of a network trained on CIFAR10. We manipulate this network to produce CKA maps which: (1) maximizes the CKA similarity between the 1st and last layer, (2) maximizes the CKA similarity between all layers, and (3) minimizes the CKA similarity between all layers. In cases (1) and (2), the network experiences only a slight loss in performance, which counters previous findings by achieving a strong CKA similarity between early and late layers. We find similar results in the kernel CKA case.

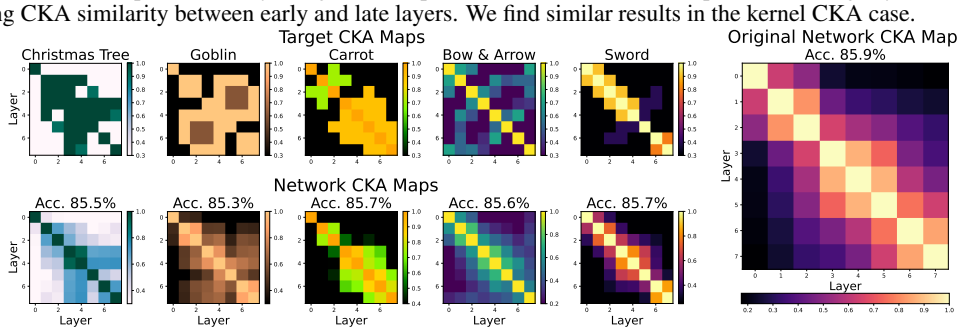

Figure 6: The comical target CKA maps (first row) are used as the objective for the CKA map loss in Eq. 3, while prioritizing network performance (small tolerance for changes in accuracy δ_{acc}). The second row shows the CKA map produced by the network.

by the last layer of the network. Therefore a meaningful way in which two sets of representations can be "similar" in practice is if they are linearly separable by the same hyperplanes in parameter space, with the same margins. Given X, the network's internal representations of 10k training images at the last layer before the output we can use an SVM classifier to extract the hyperplanes in parameter space which best separate the data (with approx. 91% success rate). We then create Y by translating a subset of the representations in a direction which won't cross these hyperplanes, and won't affect the linear separability of the representations. We plot the CKA values between X and Y according to the translation distance in Fig. 4b. The CKA values quickly drop to 0, despite the existence of a linear classifier that can classify both sets of representations into the correct classes with $> 90\%$ accuracy. In Fig. 4b we also examine linear CKA's sensitivity to outliers. Plotted are the CKA values between the set of training image representations and the same representations but with a single point being translated from its original location. While the translation distance needed to achieve low CKA values is relatively high, the fact that the position of a *single* point out of *tens of thousands* can so drastically influence the CKA value raises doubts about CKA's reliability as a similarity metric.

We note that our main theoretical results, namely Th. 1, Cor. 2 and Cor. 3 are entirely class and direction agnostic. The empirical results presented in this section are simply examples that we deemed particularly important in the context of ML but the same results would hold with any subset of the representations and translation direction, even randomly chosen ones. This is important since the application of CKA is not restricted to cases where labels are available, for example it can also be used in unsupervised learning settings (Grigg et al., 2021). Furthermore, the subset translations presented here were added manually to be able to run the experiments in a controlled fashion but these transformations can naturally occur in ANNs, as discussed in Sec. 3, and one wouldn't necessarily know that they have occurred. We also run experiments to evaluate CKA sensitivity to invertible linear transformations, see the Appendix for justification and results.

4.3 Explicitly Optimizing the CKA Map

The CKA map, commonly used to analyze network architectures (Kornblith et al., 2019) and their inter-layers similarities, is a matrix M, where $M[i, j]$ is the CKA value between the activations of

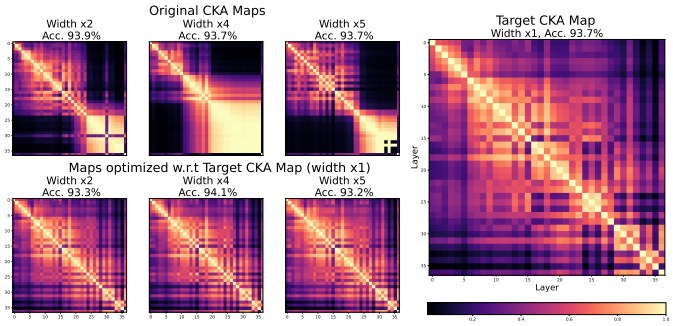

Figure 7: ResNet-34 networks of different widths and their corresponding CKA Maps are modified to produce CKA maps of thin networks. Top row **Original** shows the unaltered CKA map of the networks derived from "normal" training. **Optimized** shows the CKA map of the networks after their map is optimized to mimic the thin net *target* CKA map. See Appendix for more networks.

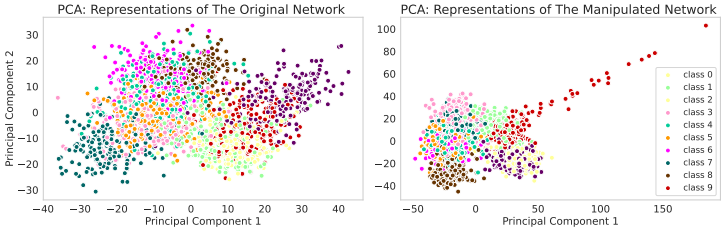

Figure 8: PCA of the networks presented in Fig. 5 before (left) and after (right) being optimized to manipulate the CKA map with Eq 3. Noticeably to achieve the objective the optimization displaces a subset of a single class.

layers i, and j of a network. In many works (Nguyen et al., 2021; Raghu et al., 2021; Nguyen et al., 2022) these maps are used explicitly to obtain insights on how different models behave, compared to one another. However, as seen in our analysis so far it is possible to manipulate the CKA similarity value, decreasing and increasing it without changing the behaviour of the model on a target task. In this section we set to directly manipulate the CKA map of a trained network f_{θ^*}, by adding the desired CKA map, M_{target}, to its optimization objective, while maintaining its original outputs via distillation loss (Hinton et al., 2015). The goal is to determine if the CKA map can be changed while keeping the model performance the same, suggesting the behaviour of the network can be maintained while changing the CKA measurements. To accomplish this we optimize f_θ over the training set (X, Y) (note however that the results are shown on the test set) via the following objective:

$$\theta^*_{new} = \text{argmin}_\theta \left(\mathcal{L}_{distill} \left(f_{\theta^*}(X), f_\theta(X) \right) + \lambda \mathcal{L}_{map} \left(M_{f_\theta(X)}, M_{target} \right) \right) \tag{3}$$

where $\mathcal{L}_{map} \left(M_{f_\theta(X)}, M_{target} \right) = \sum_{i,j} \ln \cosh \left(M[i,j]_{f_\theta(X)} - M[i,j]_{target} \right)$. The λ multiplier in Eq 3 is the weight that balances the two losses. Making λ large will favour the agreements between the target and network CKA maps over preservation of the network outputs. In our experiments λ is allowed to change dynamically at every optimization step. Using the validation set accuracy as a surrogate metric for how well the network's representations are preserved, λ is then modulated to learn maps. If the difference between the original accuracy of the network and the current validation accuracy is above a certain threshold (δ_{acc}) we scale down λ to emphasize the alignment of the network output with the outputs of f_{θ^*}, otherwise we scale it up to encourage finer agreement between the target and network CKA maps (see Appendix for the pseudo code).

Fig. 5 shows the CKA map of f_{θ^*} along with the CKA map of three scenarios we investigated: (1) maximizing the CKA similarity between the 1st and last layer, (2) maximizing the CKA similarity between all layers, and (3) minimizing the CKA similarity between all layers (for network architecture and training details see Appendix). In cases (1) and (2), the network performance is barely hindered by the manipulations of its CKA map. This is surprising and contradictory to the previous findings (Kornblith et al., 2019; Raghu et al., 2021) as it suggests that it is possible to achieve a strong CKA similarity between early and later layers of a well-trained network without noticeably changing the model behaviour. Similarly we observe that for the RBF kernel based CKA (Kornblith et al., 2019) we can obtain manipulated results using the same procedure. The bandwidth σ for the RBF kernel CKA is set to 0.8 of the median Euclidean distance between representations (Kornblith et al., 2019). In the Appendix we also show similar analysis on other σ values. We further experiment with manipulating the CKA map of f_{θ^*} to produce a series of comical CKA maps (Fig. 6) while maintaining similar model accuracy. Although the network CKA maps seen in Fig. 6 closely resemble

their respective targets, it should be noted that we prioritized maintaining the network outputs, and ultimately its accuracy by choosing small δ_{acc}. Higher thresholds of accuracy result in stronger agreements between the target and network CKA maps at the cost of performance.

Wider networks Nguyen et al. (2021) and Nguyen et al. (2022) studied the behaviour of wider and deeper networks using CKA maps, obtaining a block structure, which was subsequently used to obtain insights. We revisit these results and investigate whether the CKA map corresponding to a wider network can be mapped to a thin network. Our results for the CIFAR10 dataset and ResNet-34 (He et al., 2016) are shown in Fig. 7 (for details on the architecture and training procedure see Appendix). We observe that the specific structures associated with wider network can be completely removed and the map can be nearly identical to the thinner model without changing the performance substantially. Results with vision transformers yield similar results (see Fig. 14 of the appendix).

Analysis of Modified Representations Our focus in Sec 4.3 has been to use optimization to achieve a desired target CKA manipulations without any explicit specification of how to perform this manipulation. We perform an analysis to obtain insights on how representations are changed by optimizing Eq 3 in Fig. 8. Here using the modified network from case (2) of Fig. 5 we compute the PCA of the *test set*'s last hidden representation (whose CKA compared to the first layer is increased). We observe that a a single class has a very noticeable set of points that are translated in a particular direction, away from the general set of classes. This mechanism of manipulating the CKA aligns with our theoretical analysis. We emphasize, that in this case it is a completely emergent behavior.

5 Discussion and Conclusion

Given the recent popularity of linear CKA as a similarity measure in deep learning, it is essential to better study its reliability and sensitivity. In this work we have presented a formal characterization of CKA's sensitivity to translations of a subset of the representations, a simple yet large class of transformations that is highly meaningful in the context of deep learning. This characterization has provided a theoretical explanation to phenomena observed in practice, namely CKA sensitivity to outliers and to directions of high variance. Our theoretical analysis also show how the CKA value between two sets of representations can diminish even if they share local structure and are linearly separable by the same hyperplanes, with the same margins. This meaningful way in which two sets of representations can be similar, as justified by classical machine learning theory and seminal deep learning results, is therefore not captured by linear CKA. We further empirically showed that CKA attributes low similarity to sets of representations that are directly linked by simple affine transformations that preserve important functional characteristics and it attributes high similarity to representations from unalike networks. Furthermore, we showed that we can manipulate CKA in networks to result in arbitrarily low/high values while preserving functional behaviour, which we use to revisit previous findings (Nguyen et al., 2021; Kornblith et al., 2019).

It is not our intention to cast doubts over linear CKA as a representation similarity measure in deep learning, or over recent work using this method. Indeed, some of the problematic transformations we identify are not necessarily encountered in many applications. However, given the popularity of this method and the exclusive way it has been applied to compare representations in recent years, we believe it is necessary to better understand its sensitivities and the ways in which it can be manipulated. Our results call for caution when leveraging linear CKA, as well as other representations similarity measures, and especially when the procedure used to produce the model is not known, consistent, or controlled. An example of such a scenario is the increasingly popular use of open-sourced pre-trained models. Such measures are trying to condense a large amount of rich geometrical information from many high-dimensional representations into a single scalar $\in [0, 1]$. Significant work is still required to understand these methods in the context of deep learning representation comparison, what notion of similarity each of them concretely measures, to what transformations each of them are sensitive or invariant and how this all relates to the functional behaviour of the models being analyzed.

In the meantime, as an alternative to using solely linear CKA, we deem it prudent to utilize multiple similarity measures when comparing neural representations and to try relating the results to the specific notion of "similarity" each measure is trying to quantify. Comparison methods that are straightforward to interpret or which are linked to well understood and simple theoretical properties such as solving the orthogonal Procrustes problem (Ding et al., 2021; Williams et al., 2021) or comparing the sparsity of neural activations (Kornblith et al., 2021) can be a powerful addition to any similarity analysis. Further, data visualizations methods can potentially help to better understand the structure of neural representations in certain scenarios (e.g. Nguyen et al. (2019); Gigante et al. (2019); Horoi et al. (2020); Recanatesi et al. (2021)).